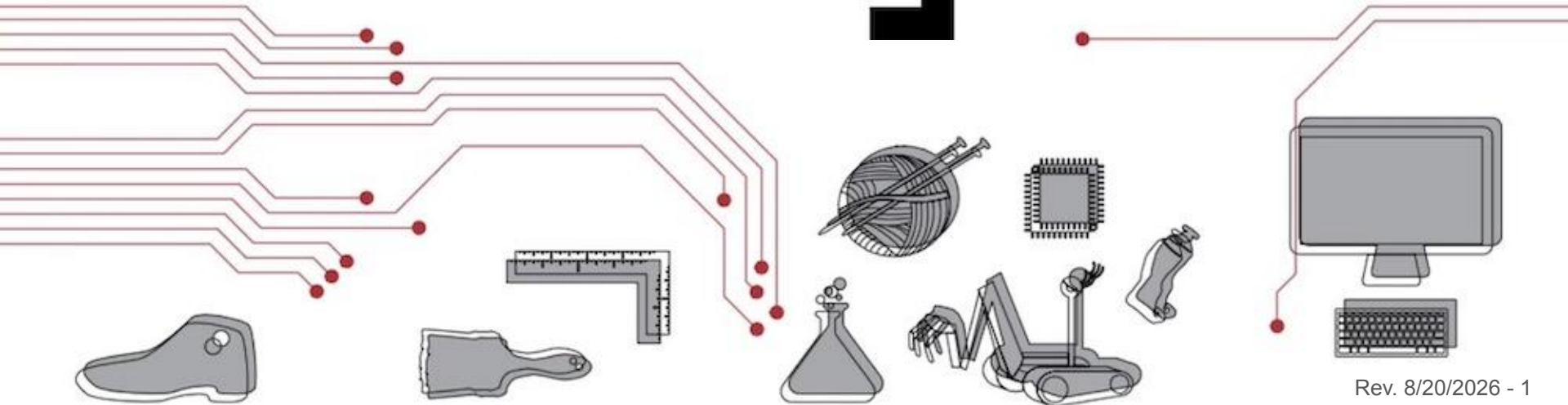


Electronics for Makers: Fundamentals & Microcontrollers

Paul Sakion



Welcome

- What projects do you want to explore?
- Experience with programming?
- Future areas of interest?
- Me: EE degree specializing in the measurement of sound and vibration
- Interests: Microcontrollers, sensors, motors, batteries, internet of things, robotics, 3D printers, etc.
- Introduction: The self balancing robot and the grunt-bot

Class Overview

- Modern electronics is more about using components: microcontrollers, smart sensors, motors, LEDs, displays, radios (WiFi, Bluetooth)
- Even knowing what to search for can be difficult: ESP32, ESP8266, Arduino, Raspberry Pi, NeoPixels, LoRa, MicroPython
- This class is an introduction to microcontrollers, inputs (sensors), and outputs
- What motor should I use? DC, AC, stepper, servos?
- Which battery should I use? Size, type, charge/discharge speed
- Richard Feynman: Just learn what is needed to understand/solve problems

Basic Safety

- One hand rule, buddy rule, capacitors, hot components
- Static grounding / Power ratings / Choosing [wire size](#) / [Batteries](#)

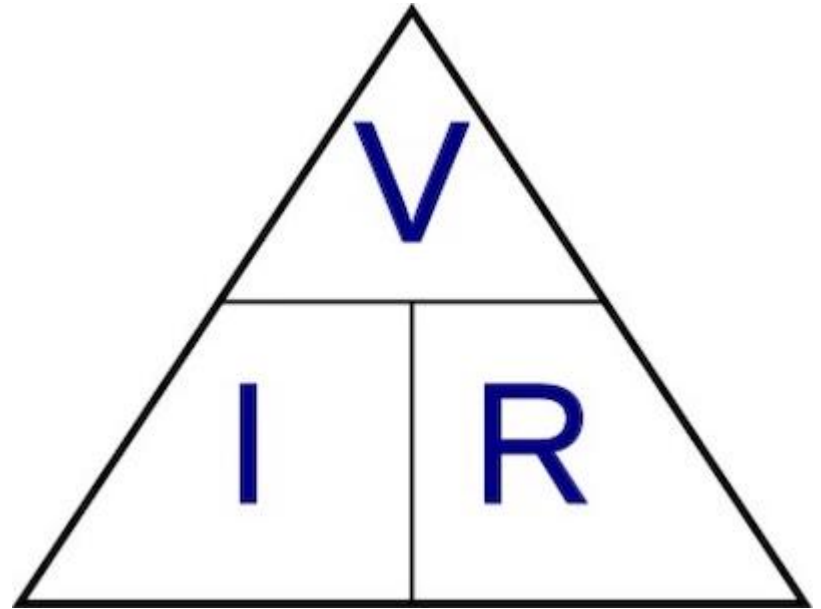


Voltage

- Electrical equivalent of pressure (think garden hose)
- Measured between points (like water pressure or distance)
- Two different types: Direct and Alternating
- Alternating Current: Power outlet, AC power adapter
- Direct Current: Battery, [Plug-in adapter \(AC to DC\)](#), USB ports, DC supply

Voltage, Current, and Resistance

- Voltage = Current x Resistance
- Voltage is electrical pressure
- Current is the flow in amps (I)
- Resistance is the friction in ohms
- 5 volts = 1 amp x 5 ohms
- (Used when calculating LED current)

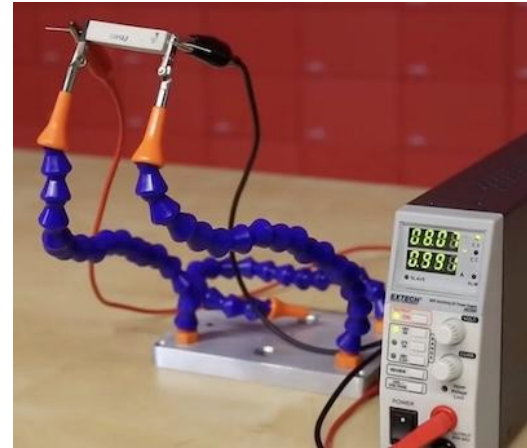


Power: Watts ($P = V \times I$)

- Work done by the flow of electrons (amps) by pressure (voltage)
- 100W incandescent vs. 10W LED
- 1500W hair dryer (120V / 15A circuit)
- Power rating of components
- An excellent [explanation video](#)



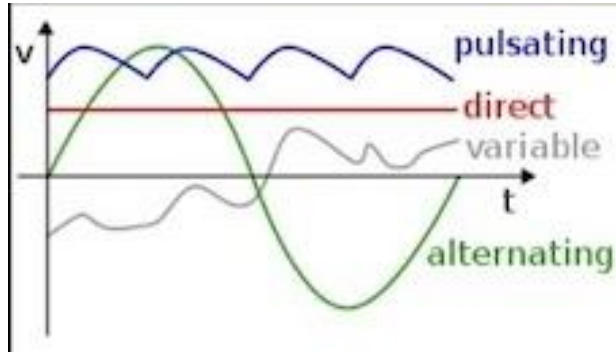
[Watt Meter](#)



[Spark Fun Video](#)

Frequency

- When voltage changes over time: Hertz or cycles/second
- Our wall sockets are 60 Hz / 120v (thanks to Tesla/Edison)
- Computer microprocessor clock ~ 4.5 GHz (Mac M4)
- Electromagnetic: 2.4 GHz (microwaves) / 5 GHz / 300 GHz
- Wavelength - $10^8 \text{ (c)} / \text{frequency (mmwaves)} = 300 \text{ GHz}$

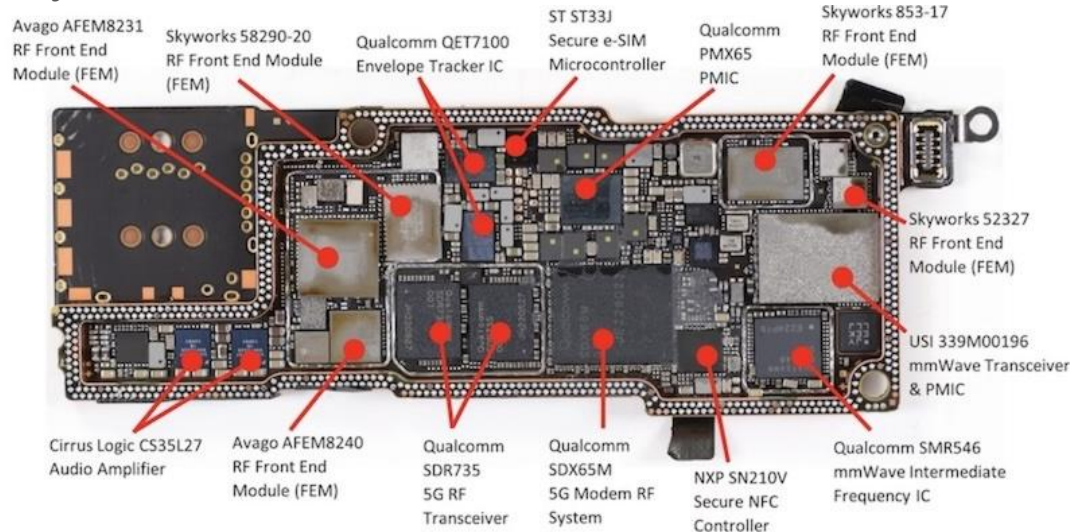


Modern Electronics

- Microcontrollers, breadboards, LEDs, sensors
- Motors (DC, AC, stepper, servos), displays, WiFi, Bluetooth
- Simulation [software](#) like [CircuitLab](#) and [TinkerCad](#)
- Demo of Multimeter and a [great video](#)
- Demo of Power Supply and [a tutorial](#)
- Explanation of oscilloscope [and another video](#)
- The SparkFun [Youtube channel](#) is awesome

Circuit Boards

- All those little specks are electronic parts
- Resistors, capacitors, inductors, integrated circuits
- Connected by wires embedded in the circuit board



Microcontrollers

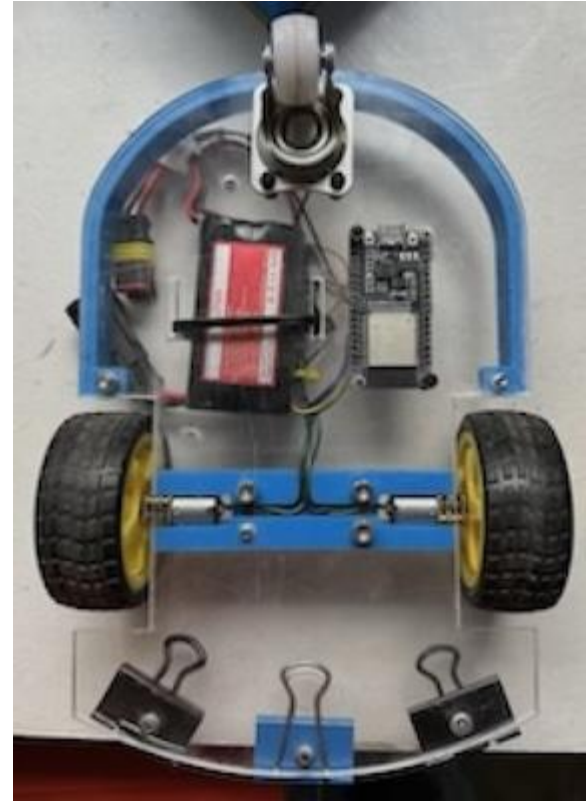
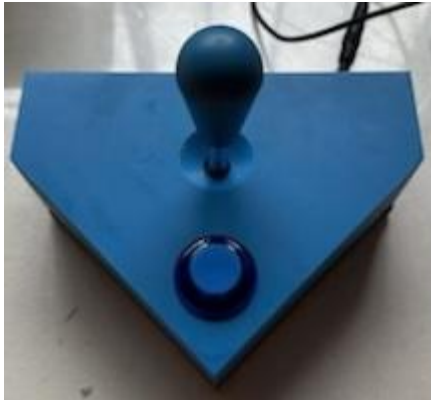
- [Arduino](#) / [ESP32](#) - Open-source hardware / software electronics platforms intended for anyone making interactive projects (Arduino [cookbook](#))
- [Raspberry Pi](#) - A credit card sized computer that can be used to learn coding and to build [electronics projects](#), also works like a desktop PC ([beginner's kit](#) / [cookbook](#) / [Sparkfun guide](#))
- [Micro:bit](#) - An open-source hardware designed for education. It has a processor, several sensors, Bluetooth / USB, 25 LEDs, two buttons, and can be powered by either USB or an external battery pack
- Video explaining many of the [common types](#)

Inputs and Outputs

- Sensors - [Thousands of types](#) including temperature, pressure, humidity, light, microphones, cameras, movement, speed, distance, switches
- Indicators - LEDs, buzzers, speakers, displays
- DC Motors - Servos (airplane flaps / car steering), Stepper (precise movement), Brushed DC (requires motor controller called an ESC), Brushless DC (drones, requires an ESC Electronic Speed Controller)
- ESC is an Electronic Speed Controller, a specialized board controlled by a PWM signal (Pulse Width Modulation)

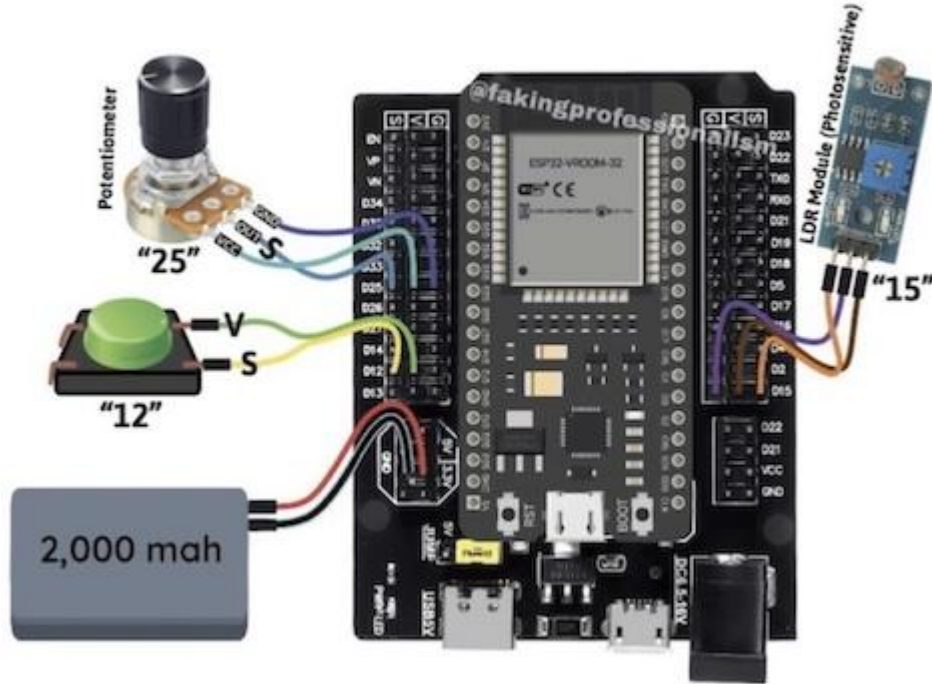
Andrew's Car

- A microcontroller, motors, motor driver, battery, voltage boost converter (3.7 volts to 12 volts)
- 3D printed and laser cut body



Inputs

- Button (digital)
- Battery (supply voltage or measure level)
- Light level sensor (analog)
- Potentiometer or “pot” (analog variable resistor)



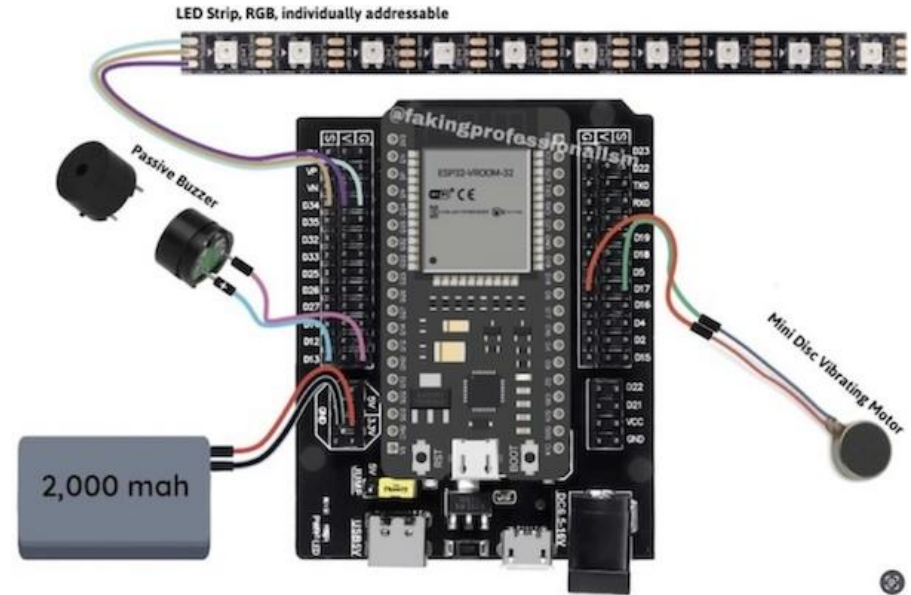
More About Analog Inputs

- Pushbuttons - Pull up and pull down [resistors](#)
- Voltage inputs - Analog to digital [converters](#) (ADC), battery voltages (3.3v)
- Higher voltages - Requires a [voltage dividers](#)
- Current measurement - Current [shunt](#) (a precision, low-resistance resistor)
- Potentiometers or “Pot” (type of [voltage divider](#))
- ESP32: [issue](#) with reading 3.3v vs 3.2v
- Some devices have better ADCs



Outputs

- Passive buzzer (low quality speaker)
- Vibrating motor (PWM)
- NeoPixel (WS2812) LED strip (digital, one output for all)
- Battery (but these pins can be voltage output as well)



More About Analog Outputs

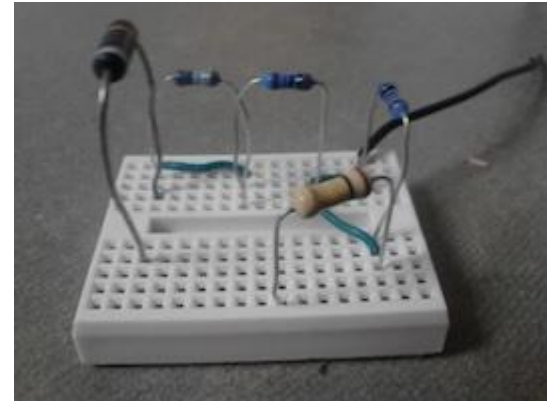
- LED - One color for each LED and needs a current limiting resistor as explained in [this guide](#)
- DAC (Digital to Analog Converter) - For a beeper or can be used as a low quality [speaker](#)
- [PWM](#) (Pulse Width Modulation) - A common control output that can control everything from the brightness of analog LEDs to precise positioning of [servos](#) to motor speed in [motor controllers](#)

Digital Inputs and Outputs - Outputs

- “Busses” - [I2C](#) (inter-integrated circuit bus) and [SPI](#) (serial peripheral interface bus)
- Temperature / Humidity - [DHT22](#)
- Distance - [HC-SR04](#) “sonar” (time of flight) sensor
- Neopixel LEDs - Adafruit [guide](#) and a technical [guide](#)
- LCD Displays - Two line display [guide](#) (I2C)
- LED Matrix Displays - MAX7919 [guide](#)
- OLED Displays - SSD1306 [guide](#)

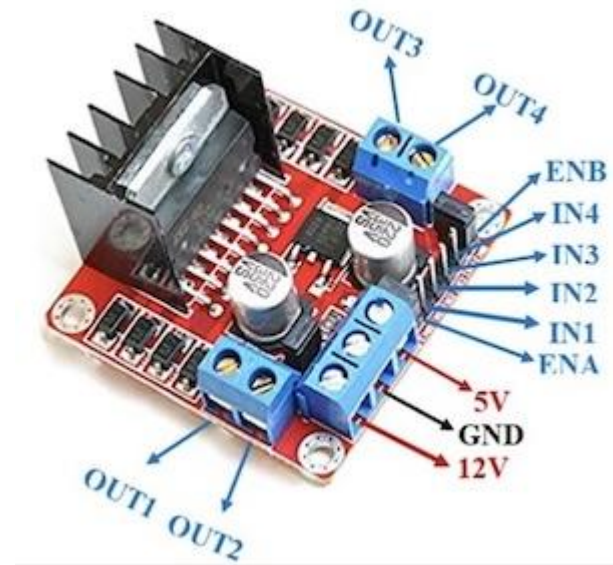
Building Circuits

- Breadboards (no soldering)
- [Breakout boards](#) - easier access to each pin
- Printed Circuit Boards (PCBs) - As low as [\\$2 for 5](#) but requires design skills ([KiCad](#)) and soldering skills
- An expert level [example video](#)



Troubleshooting

- Requires a systematic approach
- Break down and simplify the problem by isolating parts
- Review the spec sheet, confirm inputs, confirm outputs
- Test power to motor then test motor then test connections
- There's a [whole book](#) on it



MicroBlocks Programming

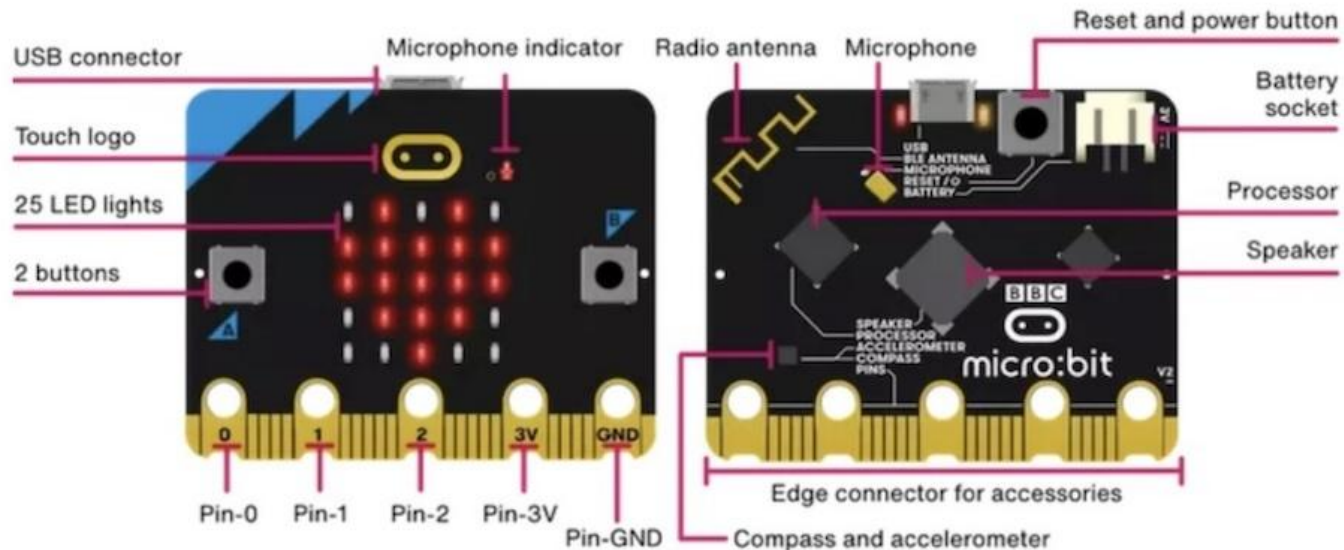
- Easy way to [get started](#) with real-time programming
- Update firmware first with detailed instructions on the [Make Wiki](#)
- Command, reporter, “C,” “hat” blocks and other toolboxes
- Using Libraries and the [Help System](#)
- The “Hello World” of microcontrollers - Blink!
- Learn about reporter blocks with the tilt demo

MicroBlocks Programming

- Load LED DISPLAY library
- Display a letter/number on LED
- Display longer text and numbers on LED
- Display icon pictures selected from the menu
- Program button A / B: display text, picture, animation
- Discover the ENCODED information - everything is a number

Micro:bit Microcontroller

- Micro:bit [Overview](#) - Open-source hardware with a processor, sensors, Bluetooth/USB, 25 LEDs, two buttons, microphone, USB or battery power



PicoBricks

- An electronic development board that includes several sensor modules
- OLED display, Neopixels, temperature, presence, relay, keyboard, motor/servo drivers, communications, gesture, light, and a potentiometer



PicoBricks Exploration

- Explore hardware features: Display, RGB LEDs, Temp/Humidity sensor, Passive IR (PIR) sensor, Relay, Motor module, Gesture sensor, LDR Light sensor, Potentiometer, Touch key sensor
- Learn about basic operational principles and use of the modules
- Display: bitmap images, print information and graphics
- RGB LEDS: how colors are made
- Temp/humidity sensor: monitor the environment, activate AC/Heating/Watering (greenhouse / home automation applications)

PicoBricks Activities

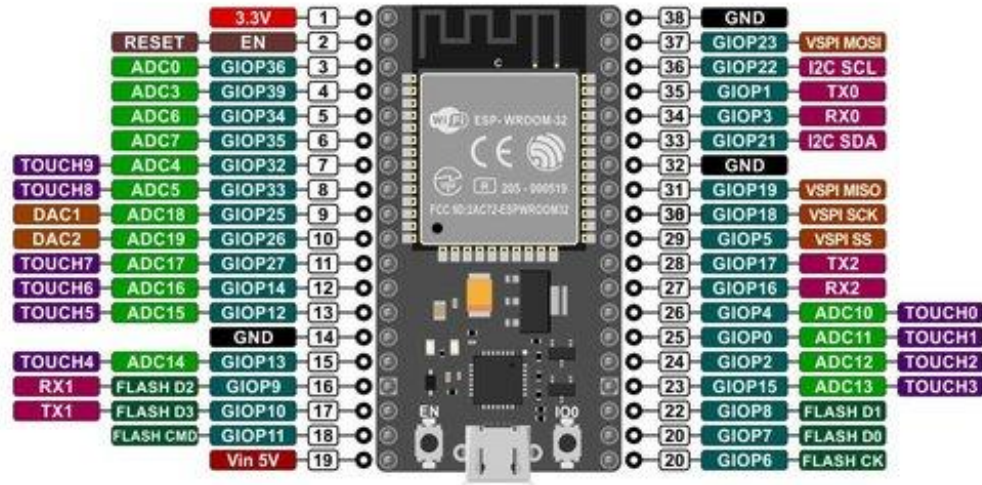
- PIR sensor: detect motion in area, use to build alarm systems, monitor activity
- Relay: turn power on/off, use to control devices, motors, lights, appliances
- Motor module: DC / Servos, cars, robots, open/close entrances
- Gesture sensor: detect hand motions nearby, use to control activities remotely
- LDR Light sensor: detect light levels, use to control lights or window shades
- Potentiometer: turn to change, use for input for sound, light, temp, etc control
- Touch Key sensor: activates when touch is detected, use for control of anything you can think of or play music/games

Advanced Project with PicoBricks

- Let's make a letter and display it on the screen: first with micro:bit, then on OLED display
- OLED display: text in various sizes, geometric shapes
- What's with all the blinking going on the OLED display, how do we get rid of it?
- Program the pot to control OLED brightness using backlight
- RGB colors (0-255): primary colors and combination colors
- Program NEOPixels for desired patterns, colors, movements
- Program to display temp/humidity on the OLED screen
- Explore displaying values in multiple lines, in various sizes
- Program the PIR sensor to alert for motion and sound an alarm
- Motor module: make motors and servos move
- Program hand movements to control micro:bit or other modules, display pictures
- Program LDR light sensor to dim/brighten OLED display
- Program to plot potentiometer values, rescale values to change (for example): size, volume, brightness, etc.
- Use a touch key sensor to play musical notes and display the note names on the OLED Display. Make the note play duration controlled by the user (HINT: need sensor input)
- Display multiple sensor values on the OLED and update them in real-time Eg: temp, potentiometer, Light sensor, etc.
- Design and implement a system with sensor detection and corresponding action. Eg: Alarm system, Appliance use automation, turn on/off motors, etc.
- Design a mini broadcast system based on Radio that transmits messages, pictures, sensor values over different channels. Users will tune into the channel and receive the information.
- Use joystick to remotely control a servo over Wifi (requires two micro:bit controllers)

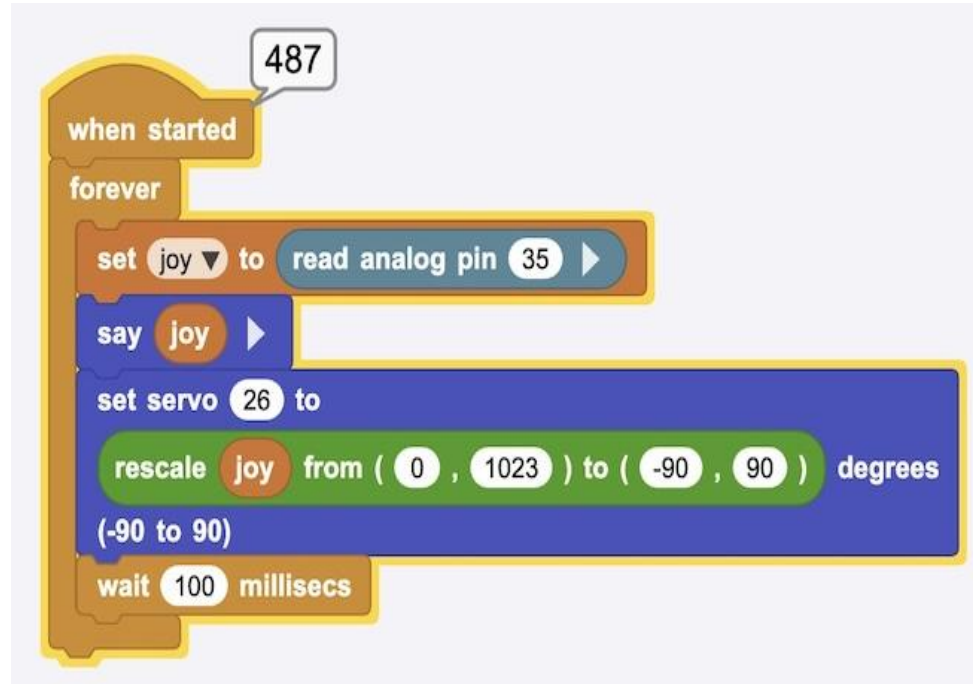
ESP32 Tips and Tricks

- Choose pins that are typically safe to use by referencing [this tutorial](#)
- Input/output pins: 4, 5, 13, 14, 18, 19, 21, 22, 23, 25, 26, 27, 32, 33
- [More ESP32s](#) / [Video](#) of options
- Digital to analog pins: 25 and 26
- Input only: 34-39 (DHT11 issue)
- I2C SDA/SCL: Pin 21 and 22
- WiFi: don't use 12-15, 25-27
- More tips on the [Wiki Page](#)



Joystick and Servo Demo

- Can connect small servos to the ESP32 directly (needs 5v power)
- Connect joystick to pin 35: 3.3v to “+5v” pin, GND, VrX or VrY
- Output of Say (Analog Pin 36) should be around 500
- Connect servo to pin 26: Red (5v), Brown (Gnd), Orange (Signal)
- Hands-on is the [best way to learn](#)



Cutebot Pro Car Kit

- A programmable robot with a 4-way infrared line-following sensor, an encoder motor, the LED rainbow light, an ultrasonic sensor, and other devices



CuteBot Overview

- Hardware features: Chassis, power switch, battery, encoder motors, Micro:bit slot, distance sensor, IR sensor, headlights (LEDs), NEOPixel Lights, line tracking sensors, calibration button, Expansion ports
- Basics of Robotics 3C's: communication, command, and control.
- Communication: IR Remote control, Radio Control
- Command: manipulating the car using library commands that are preprogrammed, getting information from the sensors
- Control: maneuvering the car in various ways: by motor power %, by distance, by wheel rotation, programming responses to sensor data, acting on external events.
- What is a Control LOOP? What is programmed versus continuous control?
- What is a Line tracking sensor, how does it work?
- What is an Encoder and why do we need it?
- What are minimal car controls that we need to successfully maneuver the car?
- What is an IR Remote Controller, how does it work?

Cutebot Projects

- Program car movements: forward, backward, turns in place
- Program turns by rotation angles forward and backwards
- Program car movements using sensor values: distance, speed, obstacles
- Program a script to detect lines
- Program a control script using IR Remote for car control
- Program displays for telemetry information display
- Program obstacle avoidance using the distance sensor
- Program the car to follow the track
- Program a control script using radio for car control
- Enhance car movements by using the headlights, NEOPixels, and sounds

Other Programming Options: MicroPython

- Python designed to work under limitations like those found in microcontrollers
- Includes modules to access low-level hardware like GPIO
- Several steps required before it can be used with a microcontroller (upload base firmware, download and use the Thonny IDE, etc.)
- Tutorials are [available](#) and AI can be used to write code

Other Programming Options: Arduino IDE

- Web based Arduino [code editor](#) (IDE) where you can write code, access libraries, and upload to board (ESP32 and other boards as well)
- Tutorials are [available](#) and a [language reference](#)
- The Arduino [Cookbook](#)
- Many kits are [available](#): For less than \$30 get ESP32, temperature/humidity, distance, joystick, infrared, and other misc parts
- More kits are available on the Make Electronics [Wiki page](#)

Analog Components

- Capacitor - Stores energy in an electric field (like a tiny tank). Two conductive plates separated by an insulator.
- Inductor - Stores energy in a magnetic field (like a tiny water wheel connected to a mass). A coil of wire.
- Resistor - Resists the flow of electrons like a thinner piece of pipe. A wire with resistive material (carbon, etc.)
- Lots of [videos online](#)

Other Components and Circuits

- Diode - Allows current to flow only in one direction
- Transformer - Two inductors coupled by a magnetic field
- Transistor - Semiconductor used to amplify or switch
- Integrated Circuits (ICs) - Combines transistors, resistors, and capacitors
- Series and parallel [circuits](#)
- Analog Circuits - Current or voltage vary continuously with time
- Digital Circuit - Voltages are assigned to discrete values (ex. 0=0v / 1=3.3 or 5v) to represent logical and numeric values

USB-C Power

- Unlike USB-A the voltage can vary and be negotiated with a trigger board
- Standard voltage steps are 5 V, 9 V, 15 V, 20 V, and 28V
- Formal name is USB-C Power Delivery
- Qualcomm has a proprietary standard called QC
- Experimental boards: [HUSB238](#) and [ZY12PDN](#)
- A more complete [guide](#) and a [video](#)



Solar Powered Systems

- Several factors - wattage, temperature, type, orientation...
- Mono-crystalline - higher efficiency / cost than poly
- 400-watt panel can produce about 1,600 watts / day
- 400-700 watts - can power a range of small appliances
- 1400 watts - Running a small air conditioner for 12 hours
- [Semi DIY \\$450](#) - 200 Watts 12 Volt/24 Volt Mono-crystalline Solar Panels with 30A PWM Charge Controller and 100aH LiFePO4 Battery
- Solar “[Generator](#)” System \$700 - 200W Solar Panel, 1070 Wh LiFePO4 Battery, 1500W AC/100W USB-C Output

Batteries

- Lithium Ion (Li-ion) - Good general rechargeable, usually cylindrical. Found in phones, laptops, flashlights, etc.
- Lithium Polymer (LiPo) - Higher discharge rate, usually rectangular or pouch shaped
- Lithium Iron Phosphate (LiFePO₄) - Heavier, 0%-100% every day, 2000+ charge cycles (vs 500), good for solar battery systems

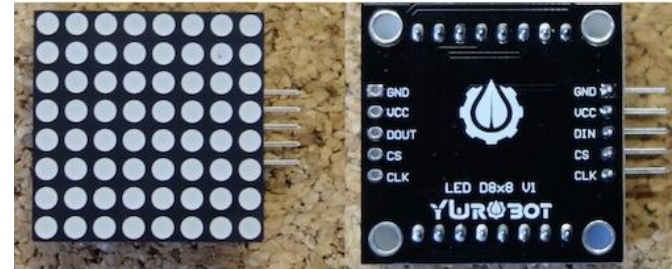
Basis	Lithium Ion	Lithium Polymer
Ageing	Loses actual charging capacity over time	Retains charging capacity better than Lithium =ion
Energy Density	High Energy density	Low as compared to lithium ion
Conversion Rate	The capacity to convert battery into actual power 85-95%	75- 85%
Safety	More Volatile as compared to lithium polymer	More safety. Less chance of explosion
Cost	Cheaper	Slightly Expensive(+30%)
Weight	Heavier	Light Weight
Charging duration	Longer Charge	Comparatively Shorter

Light Emitting Diodes (LEDs)

- LED - One color for each LED and needs a [current limiting resistor](#)
- Analog LEDs are very inexpensive (100 for \$5) - [Guide](#)
- WS2812B / “NeoPixels” - Digital so one output for many (hundreds!)
- Requires microcontroller with 5 volt logic or a “[level shifter](#)”
- Many colors, shapes, and sizes - including [wearable](#)
- [Super guide](#) / [Technical guide](#)
- Adafruit wearable [kits](#) (and [here](#) and [here](#))

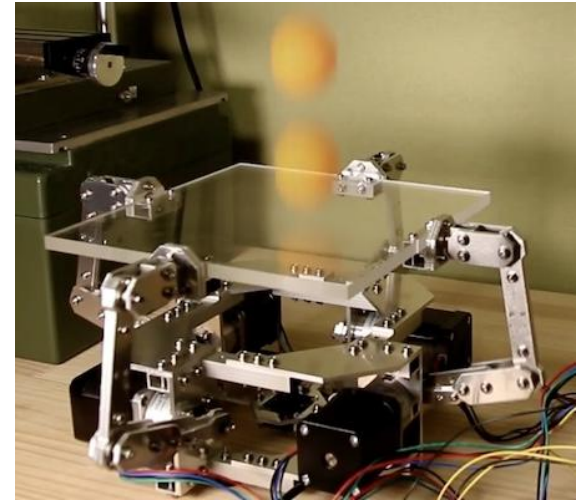
LED Control and Other Microcontrollers

- Dedicated system to use an ESP32 via a web server/ smartphone app to control NeoPixels (aka WS2812) - WiFi only (no Bluetooth)
- WLED project [website](#). Installation [website](#)
- Circuit Playground “[Ring](#)” with sensors, LEDs, BLE ([guide](#))
- LCD Displays - Two line display [guide](#) (I2C)
- LED Displays - MAX7919 [guide](#)
- OLED Displays - SSD1306 [guide](#)



Miscellaneous Interesting Devices / Projects

- Noods - Flexible LEDs “[Noodles](#)”
- Special Purpose Controllers - [Proffieboard](#) for lightsabers
- Robot [kit](#) or maybe a 3D printed self balancing [robot](#)
- Retro gaming [system](#) / Musical [instrument](#)
- DIY [weather station](#)
- Computer [vision](#) / the “[bouncer](#)”



Where to Go From Here

- Sparkfun [Engineering Essentials](#)
- Hackspace free [Maker Books](#)
- Adafruit [general guide](#) and [IoT guide](#)
- Arduino [project hub](#) (over 5,500 projects)
- [PiMyLifeUp](#) (Arduino, Linux, Electronics)
- [Circuit simulation](#)
- Digikey [Maker projects](#)
- YouTube multi-hour [electronics playlist](#)
- Links from class on the [Make Electronics Wiki](#)